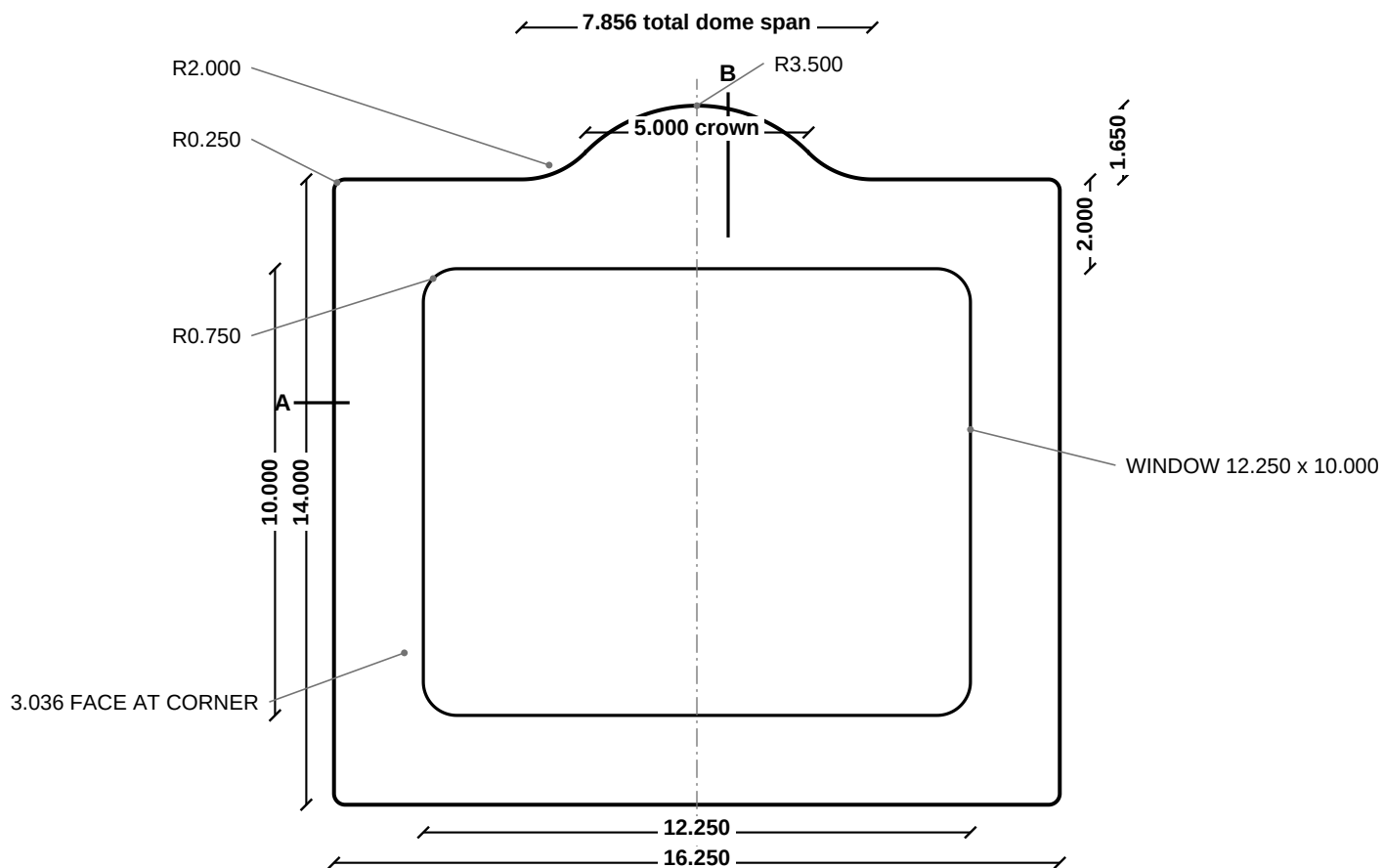


UFO PICTURE FRAME - 3D PRINT PART

SHEET 1 - FRONT VIEW

SHEET 1 OF 7

ALL DIMENSIONS IN INCHES



The crown arc, both concave joiners and the top rail must form ONE continuous tangent curve.

No corner, no kink, no flat spot at any junction. This is the most important feature of the part.

Face width is 2.000 on the straight runs and 3.036 across the diagonal at each corner. This is correct, not an error.

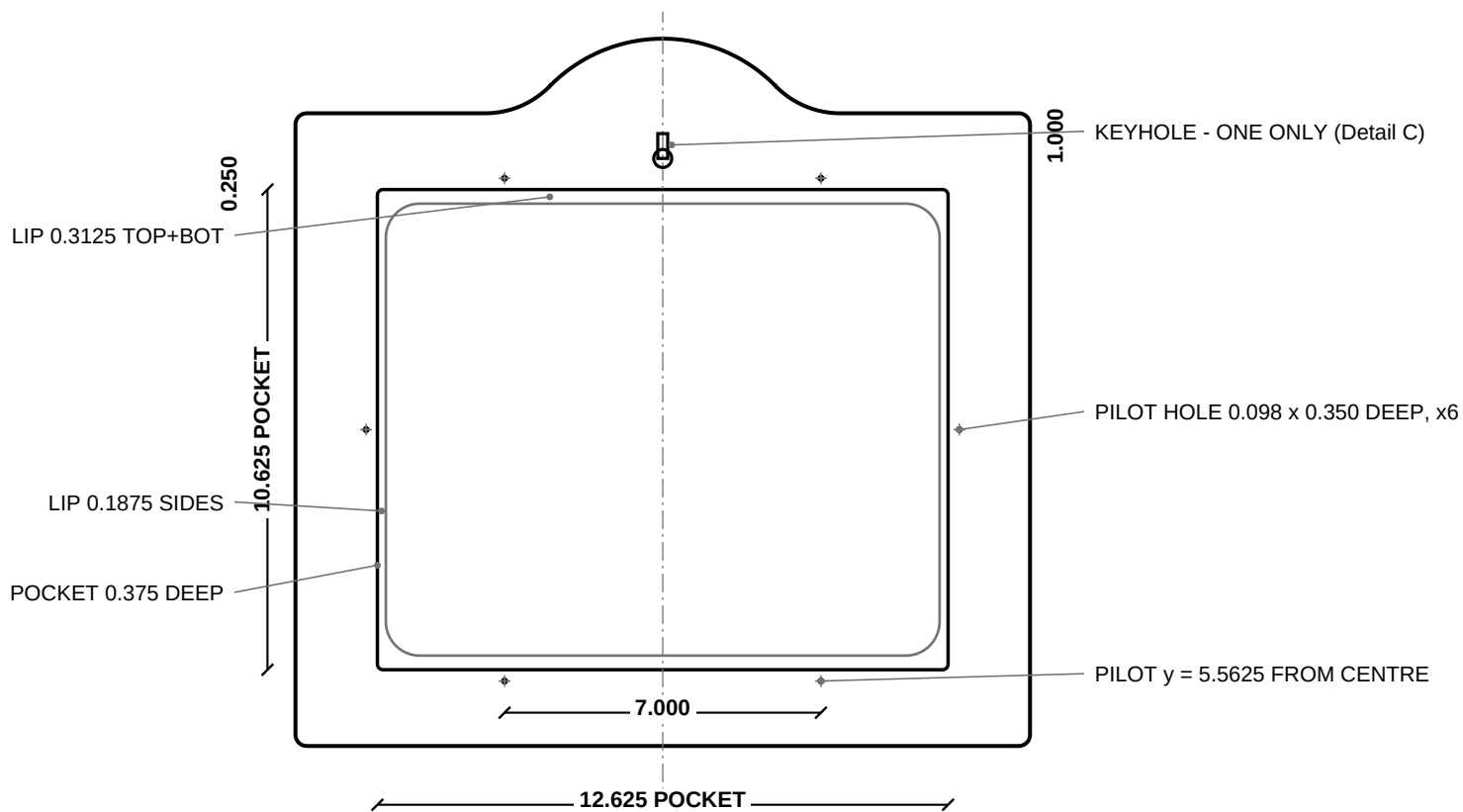
Part is symmetric about the vertical centre line. Back face features are shown on Sheet 2.

UFO PICTURE FRAME - 3D PRINT PART

SHEET 2 - BACK VIEW (the side facing the wall)

SHEET 2 OF 7

ALL DIMENSIONS IN INCHES



ALL FEATURES ON THIS SHEET ARE ON THE BACK FACE ONLY. NONE BREAK THROUGH TO THE FRONT.

Pocket floor is flat. One depth only, 0.375. Do not add any step, ledge or groove inside the pocket.

The back face of the dome is flush and flat with the back face of the frame.

The lip is 0.250 thick all round, but its WIDTH is 0.1875 on the sides and 0.3125 top and bottom.

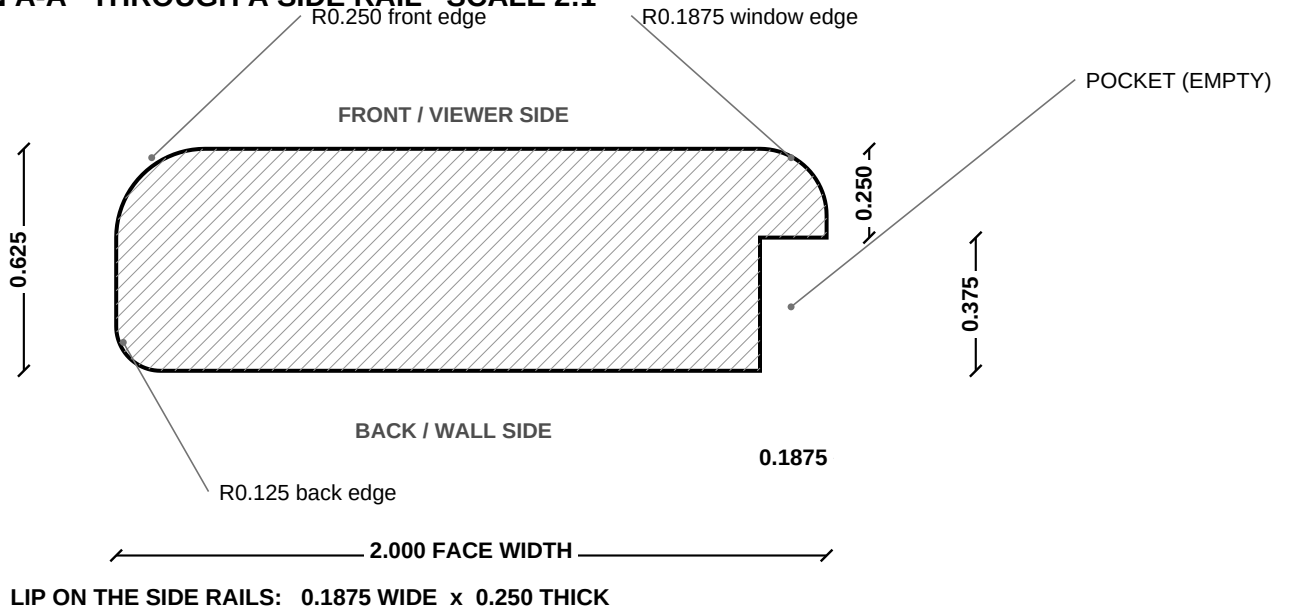
UFO PICTURE FRAME - 3D PRINT PART

SHEET 3 - SECTION A-A AND DETAIL D

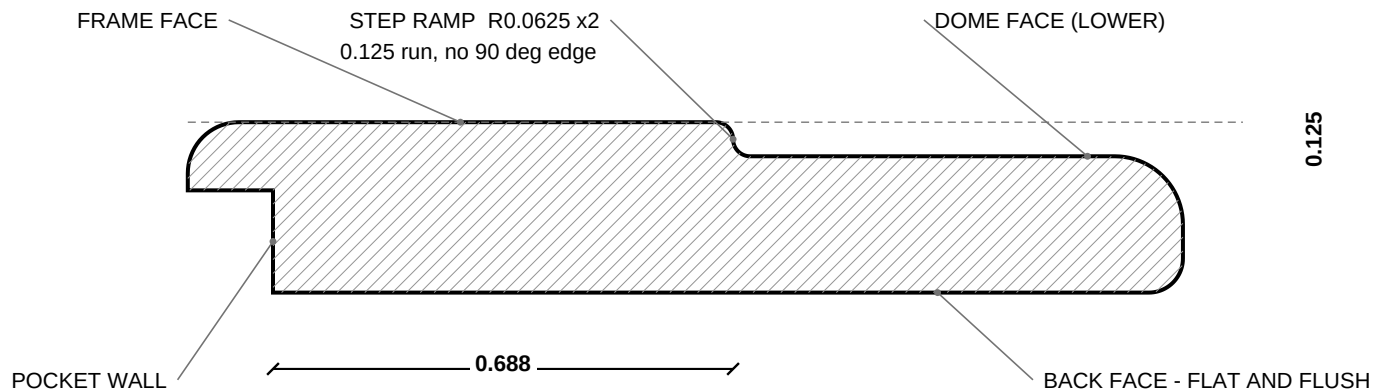
SHEET 3 OF 7

ALL DIMENSIONS IN INCHES

SECTION A-A THROUGH A SIDE RAIL SCALE 2:1



DETAIL D DOME STEP, ENLARGED SCALE 1.4:1



THE DOME FRONT FACE SITS 0.125 BEHIND THE FRAME FRONT FACE. THIS IS INTENTIONAL.

The step between them is a smooth S-shaped ramp made of two tangent radii. No flat, no corner.

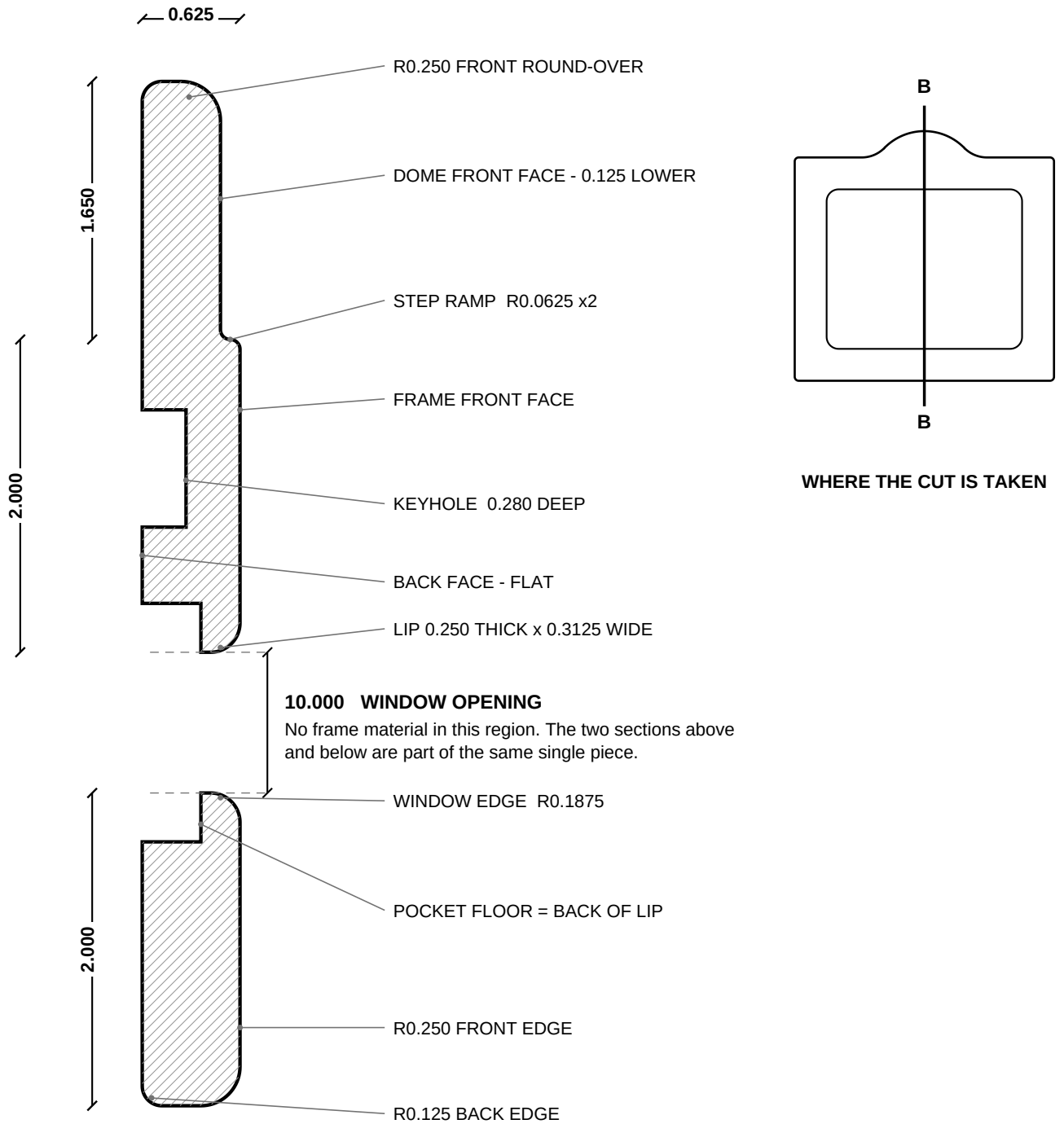
The pocket is drawn EMPTY. Do not model the glass, print or backing - they are customer parts.

UFO PICTURE FRAME - 3D PRINT PART

SHEET 4 OF 7

SHEET 4 - SECTION B-B THROUGH TOP AND BOTTOM RAILS SCALE 1:1

ALL DIMENSIONS IN INCHES



0.375 POCKET
LIP ON THE TOP AND BOTTOM RAILS: 0.3125 WIDE x 0.250 THICK

Hatched area = solid material. Everything hatched is part of ONE single piece.

Drawn at actual size. The cut passes through the centre line, so it also cuts the keyhole.

Top rail and bottom rail are identical, 2.000 each. The frame is symmetric top to bottom.

LIP WIDTH HERE IS 0.3125. On the side rails it is 0.1875 - see Section A-A. Both are correct.

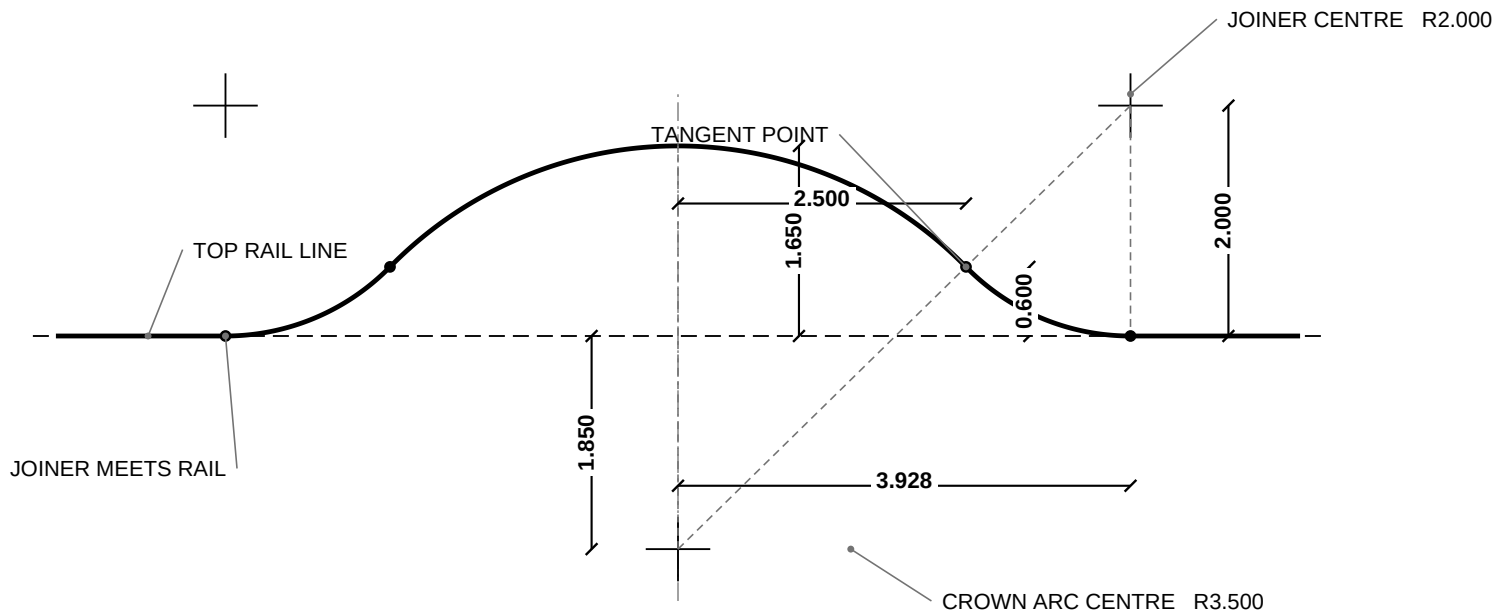
The pocket is drawn EMPTY. Glass, print and backing are supplied by the customer - see Sheet 6.

UFO PICTURE FRAME - 3D PRINT PART

SHEET 5 - DOME CONSTRUCTION GEOMETRY (build the curve from this)

SHEET 5 OF 7

ALL DIMENSIONS IN INCHES



Draw the crown arc and the two joiner arcs from the three centres above. All junctions are tangent.

Crown arc centre lies on the vertical centre line, 1.850 BELOW the top rail line.

Each joiner arc centre lies 3.928 from the centre line and 2.000 ABOVE the top rail line.

Check: distance between crown centre and joiner centre = 5.500 exactly = 3.500 + 2.000.

If that check fails, the curve is wrong. All four black dots are tangent points.

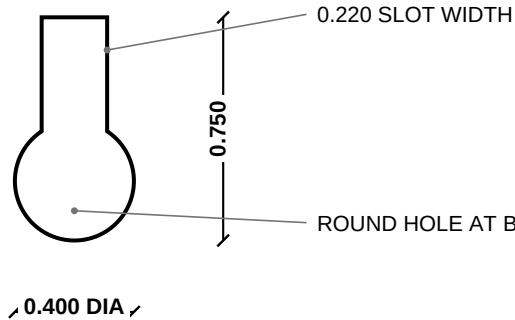
UFO PICTURE FRAME - 3D PRINT PART

SHEET 6 - DETAILS AND ASSEMBLY

SHEET 6 OF 7

ALL DIMENSIONS IN INCHES

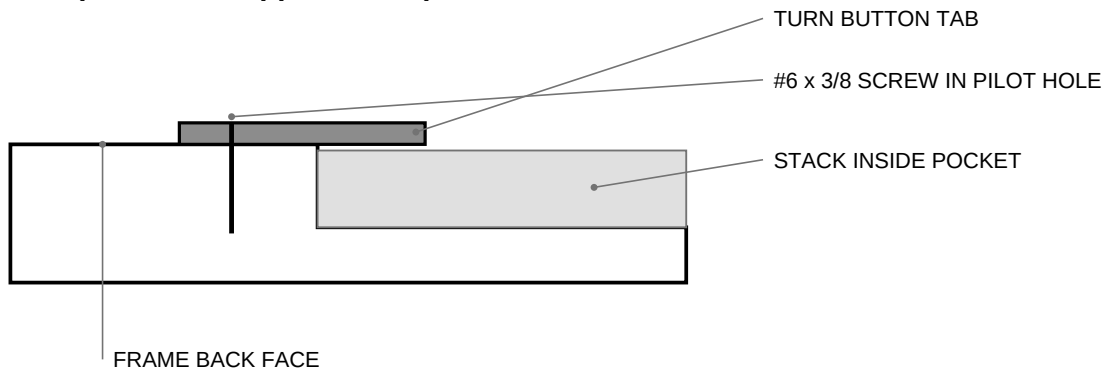
DETAIL C KEYHOLE, VIEWED FROM BACK SCALE 3:1



Pocket depth 0.280 from the back face.
Leaves 0.345 of solid material in front.
Must NOT break through to the front face.

Wall screw: #8 pan head, left standing
0.20 proud of the wall. Push the head
through the round hole, then let the
frame settle DOWN onto the slot.

TURN BUTTON (customer supplied, 6 off) SCALE 2:1

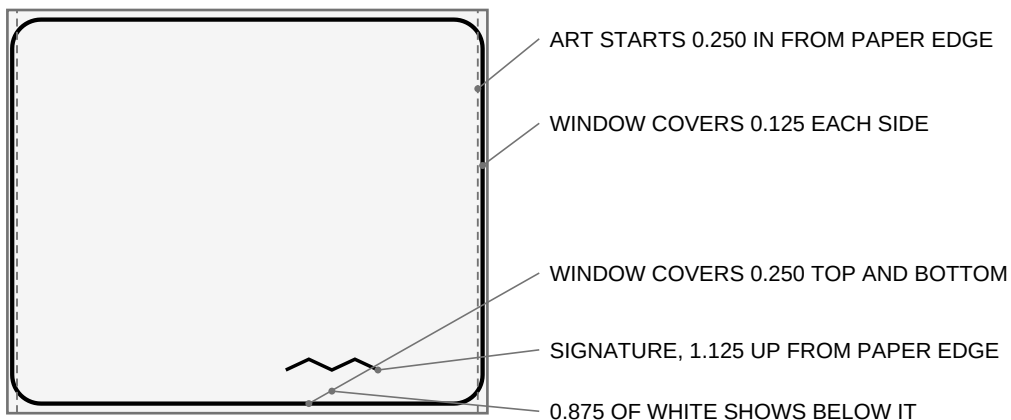


Rotate the tab over the pocket edge to lock the stack in.
Rotate it away to open the frame. Six places, evenly spaced.

ASSEMBLY ORDER (loaded from the back)

1. Lay the frame face down on a soft cloth.
2. Drop in the GLASS, 12.500 x 10.500, museum glass 2.5 mm. Clean side facing the frame.
3. Drop in the PRINT, 12.500 x 10.500, face down.
4. Drop in the BACKING, 12.500 x 10.500 x 0.250 thick foam board.
5. Fit six turn buttons and rotate each one over the backing.
6. Total stack 0.360 in a 0.375 pocket, so the backing sits just below flush and clamps tight.

ART POSITION (reference only - confirms the window size)



Grey rectangle = the paper, 12.500 x 10.500. Heavy line = the frame window. Visible white each side of the art: 0.125.

DIMENSION TABLE

Outer size	16.250 W x 14.000 H
Outer corner radius	0.250 (4 places)
Window opening	12.250 W x 10.000 H, centred
Window corner radius	0.750 (4 places)
Face width, straight runs	2.000 all four sides
Face width, at corners	3.036 across the diagonal - correct, not an error
Body depth	0.625
Dome depth	0.500 - front face sits 0.125 behind the frame front face
Dome / frame step ramp	R0.0625 convex + R0.0625 concave, 0.125 total run
Dome apex above top rail	1.650
Crown arc	R3.500, centre 1.850 below top rail line, on centre line
Concave joiner arcs	R2.000, centres 3.928 from CL and 2.000 above top rail
Crown / joiner tangent point	2.500 from CL, 0.600 above top rail
Total dome span	7.856 (crown alone 5.000)
Overall height incl. dome	15.650
Pocket, back face	12.625 W x 10.625 H x 0.375 deep, flat bottom
Pocket corner radius	0.125
Lip thickness	0.250 - same on all four rails
Lip width	0.1875 on side rails, 0.3125 on top and bottom rails
Front round-over, outer edge	0.250 continuous, includes the dome
Front round-over, window edge	0.1875
Back round-over, outer edge	0.125 continuous
Keyhole	1 only, on centre line, round hole centre 1.000 below top edge
Pilot holes	6 places, 0.098 dia x 0.350 deep, back face only
Bounding box	16.250 x 15.650 x 0.625 in / 413 x 398 x 16 mm

NOTES

1. ONE solid piece. Do NOT split the model into sections.
2. Deliver a STEP file AND an STL file. The STEP file is required.
3. Model must be parametric with a clean feature tree. Dimensions may change later.
4. No sharp 90 degree edges anywhere on the outside of the part.
5. Crown arc + both concave joiners + top rail = ONE continuous tangent curve. Highest priority.
6. Front round-over runs continuously around the entire outer edge, including over the dome.
7. Pocket is one flat-bottom recess. One depth only. No steps, ledges or grooves inside it.
8. All back face features (pocket, keyhole, pilot holes) must not break through to the front.
9. Minimum wall thickness anywhere: 0.060.
10. FDM print orientation: flat back face down on the build plate. Must print as one piece.
11. Customer will paint metallic silver. Surface finish is not critical.
12. The lip is 0.250 thick everywhere, but its WIDTH is 0.1875 on the sides and 0.3125 top and bottom.
This is deliberate. It follows from the pocket and window sizes. Do not make them equal.
13. Step ramp radius is 0.0625. This is the MAXIMUM possible on a 0.125 step.
Two fillets on a 0.125 step cannot exceed 0.0625 each. Do not increase it.
14. Send a screenshot of the finished model before delivering files.